

Day 25 Student Handout

Geometry Summer Credit | Mixed Review: Transformations, Similarity, and Trig

Name:	Date:	Period:
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Learning Targets

I can review transformations, congruence, similarity, and right-triangle trigonometry. I can choose the right tool for a mixed problem. I can explain common errors before the cumulative assessment.

Lesson 25A: Transformations, Congruence, and Similarity

Warm-Up

1.	Reflect A(6, -3) across the x-axis.
2.	Dilate B(-4, 5) by scale factor 3 about the origin.
3.	Two similar triangles have corresponding sides 8 and 20. Find the scale factor from smaller to larger.

Tool Choices

Rigid motions preserve size and shape. Dilations change size by a scale factor. Similar figures have congruent corresponding angles and proportional corresponding side lengths.

Problem	Work / notes
Point P(-2, 7) is translated by $(x+5, y-4)$, then reflected over the y-axis. Find the final image.	After translation: After reflection:
Triangle ABC is similar to triangle DEF. $AB=9$, $DE=15$, and $BC=12$. Find EF.	Scale factor: Proportion: Missing side:
A student uses SSA to prove triangle congruence. Explain why that is risky.	Valid shortcuts: Why SSA fails:

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Lesson 25B | Right-Triangle Trig Review

Right-Triangle Decision Map

Choose the Tool

Use Pythagorean theorem when two sides are known. Use special right triangles when the angles are 45-45-90 or 30-60-90. Use sine, cosine, or tangent when an acute angle and a side are involved.

Problem	Work / notes
A right triangle has legs 9 and 12. Find the hypotenuse.	Formula: Substitute: Answer:
In a right triangle, angle A=35 degrees, opposite side=10, and hypotenuse=x. Write an equation and solve for x.	Ratio: Equation: Answer:
An angle of elevation is 28 degrees. The horizontal distance is 40 ft. Find the height to the nearest tenth.	Diagram labels: Trig ratio: Answer:

4.	What is one clue that a problem needs similarity instead of trig?
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5.	What is one clue that a problem needs trig instead of Pythagorean theorem?
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